RESEARCH PAPER

Product Line Profitability and Margin Performance at Nassau Candy Distributor

A Data-Driven Descriptive Analysis of Sales, Gross Profit, Product Concentration, Regional Performance and Customer Value

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Academic Year: 2026–2027

Dataset: Nassau_Candy_Distributor.csv
Observation period: 2 January 2024 – 31 December 2025

Abstract

This study examines product-line profitability and gross-margin performance at Nassau Candy Distributor using 10,194 transaction records covering 15 products, three divisions, four regions and 5,044 customers from 2 January 2024 to 31 December 2025. The analysis uses Python and Pandas to validate records, calculate gross profit as sales minus cost, and evaluate product, division, regional, concentration and customer-level performance. Total sales were 141,783.63 dataset currency units and gross profit was 93,442.80, producing a weighted gross margin of 65.91%. The five leading products generated 95.1% of gross profit. Chocolate contributed 95.1% of gross profit, while Pacific was the leading region. Product profit HHI was 1,831. A transparent quintile-based RFM procedure is applied descriptively. Findings indicate that revenue alone is insufficient for evaluating product performance. The study is observational and gross profit is not net profit.

Keywords: Business intelligence; product profitability; gross margin; Pareto analysis; HHI; RFM segmentation; sales analytics; Python.

1. Introduction

Distribution businesses manage products with different selling prices, costs and sales volumes. Revenue alone does not show which products contribute most to gross profit. Divisions and regions can also differ in sales scale and margin. This research transforms transaction records into product, division, regional, concentration and customer measures to support evidence-based management review.

2. Problem Statement

The central problem is the limited decision value of transaction data when it is viewed only through total sales. Management needs to identify leading gross-profit products, weak-margin products, division and regional differences, product dependency and customer purchasing patterns. This study addresses these questions with reproducible descriptive calculations.

3. Objectives and Research Questions

• Audit the supplied transaction data.

• Calculate sales, cost, gross profit, weighted margin, units, orders and customer counts.

• Compare product and division profitability.

• Evaluate regional performance.

• Measure product concentration using Pareto and HHI.

• Apply reproducible RFM segmentation.

• Develop management recommendations.

Research questions: Which products contribute most to gross profit? Which have weak margins? How do divisions and regions differ? How concentrated is gross profit? What customer groups emerge from RFM measures?

4. Dataset and Data Quality

The CSV contains 10,194 rows and 18 columns. The analysis uses Order ID, Order Date, Customer ID, Product Name, Division, Region, State/Province, Sales, Cost, Gross Profit and Units. Parsed Order Dates range from 2 January 2024 to 31 December 2025.

Check	Result	Treatment
Rows / columns	10,194 × 18	All retained
Missing values	0	No imputation
Exact duplicates	0	None identified
Rows identical except Row ID	313	Retained pending source verification
Gross Profit arithmetic mismatch > 0.01	0	Recomputed as Sales – Cost

Check	Result	Treatment
Invalid dates / basic invalid values	0	No rows excluded
Ship Date	Implausible future dates	Excluded from time analysis

Order ID year tokens do not agree with Order Date; Order Date is used as the time dimension. Currency is not independently verified, so figures are labelled dataset currency units. Factory mapping is not provided and factory-level conclusions are excluded.

5. Methodology

5.1 Design and preparation

The study is quantitative, retrospective and descriptive. Order Date is parsed as dd-mm-yyyy and gross profit is recalculated as Sales minus Cost. Missingness, duplicate rows, arithmetic consistency and basic validity are checked. All 10,194 rows are retained.

5.2 Metrics

Gross Profit = Sales − Cost. Gross Margin (%) = aggregate Gross Profit ÷ aggregate Sales × 100. Revenue Contribution (%) = product Sales ÷ total Sales × 100. Profit Contribution (%) = product Gross Profit ÷ total Gross Profit × 100. Profit per Unit = Gross Profit ÷ Units. Average Order Value = total Sales ÷ distinct Order IDs. Weighted margins use aggregate totals, not simple averages.

5.3 Concentration and RFM

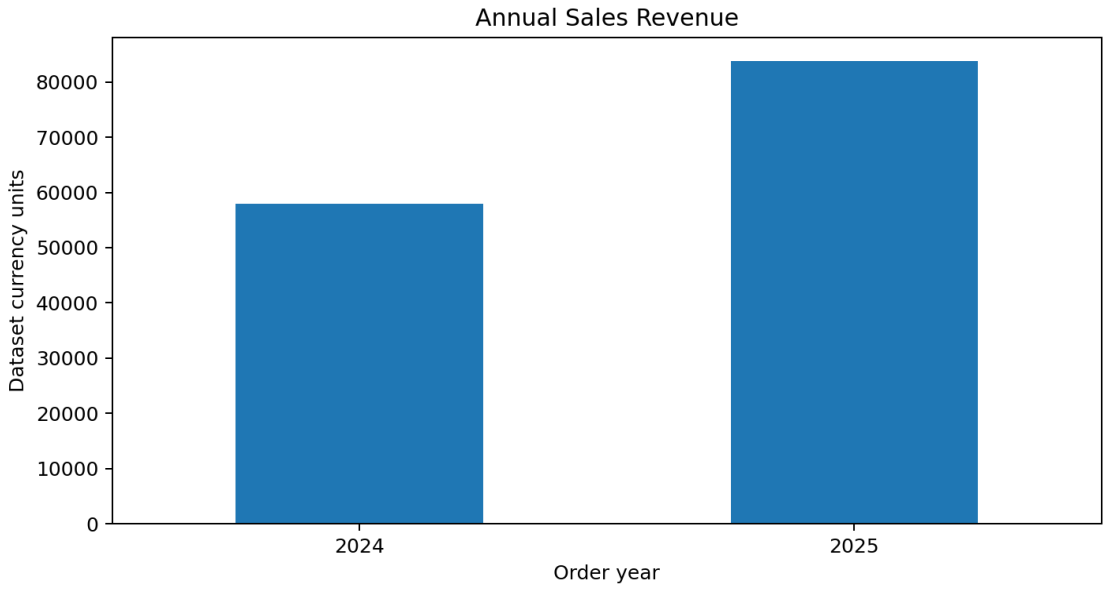
Products are ranked by gross profit and cumulative contribution is calculated. Product HHI is the sum of squared product profit-share percentages on a 0–10,000 scale. RFM recency is measured from each customer's last order to 1 January 2026; frequency is distinct Order IDs; monetary is total Sales. Quintile scores range from 1 to 5, with recency reverse-scored. Total scores 13–15 are Champions, 10–12 Loyal, 7–9 Developing and 3–6 At Risk / Low Value. These analyst-defined labels are descriptive, not predictive.

6. Results

6.1 Overall performance

The dataset contains 10,194 lines, 15 products, 8,549 orders, 5,044 customers and 38,654 units. Sales total 141,783.63, cost ('48,340.83',) and gross profit 93,442.80 dataset currency units. Weighted gross margin is 65.91%; average order value is 16.58.

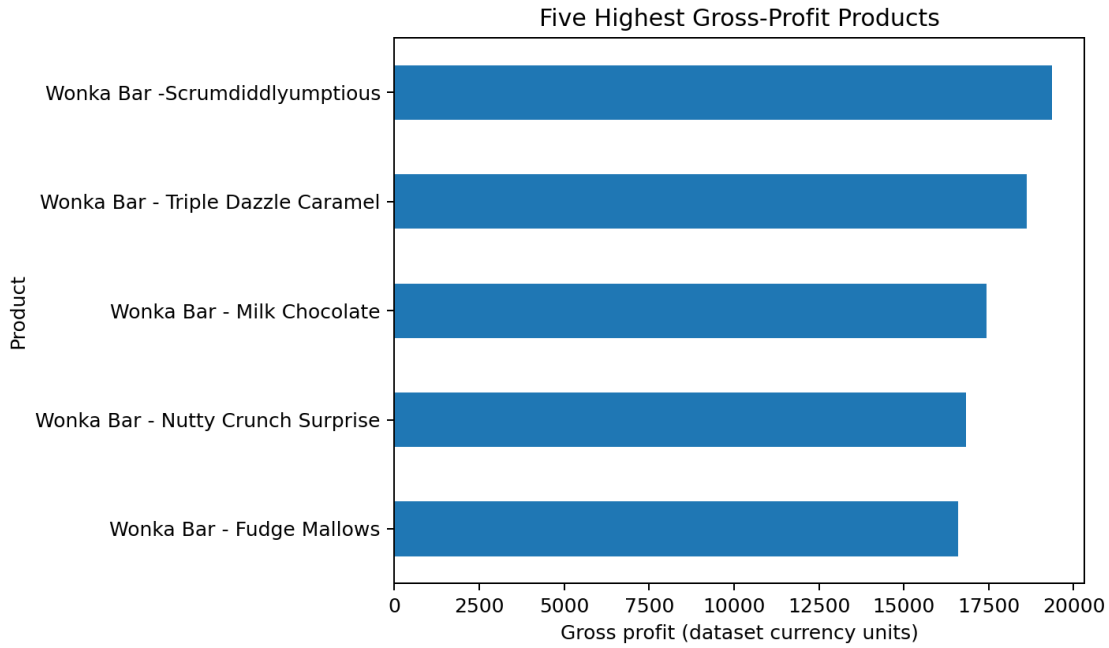
Metric	Result
Sales	141,783.63
Cost	48,340.83
Gross profit	93,442.80
Weighted gross margin	65.91%
Orders	8,549
Customers	5,044
Units	38,654
Average order value	16.58



Sales increased from 57,956.20 in 2024 to 83,827.43 in 2025, an increase of 44.6%. Gross profit increased from 38,151.43 to 55,291.37.

6.2 Product profitability

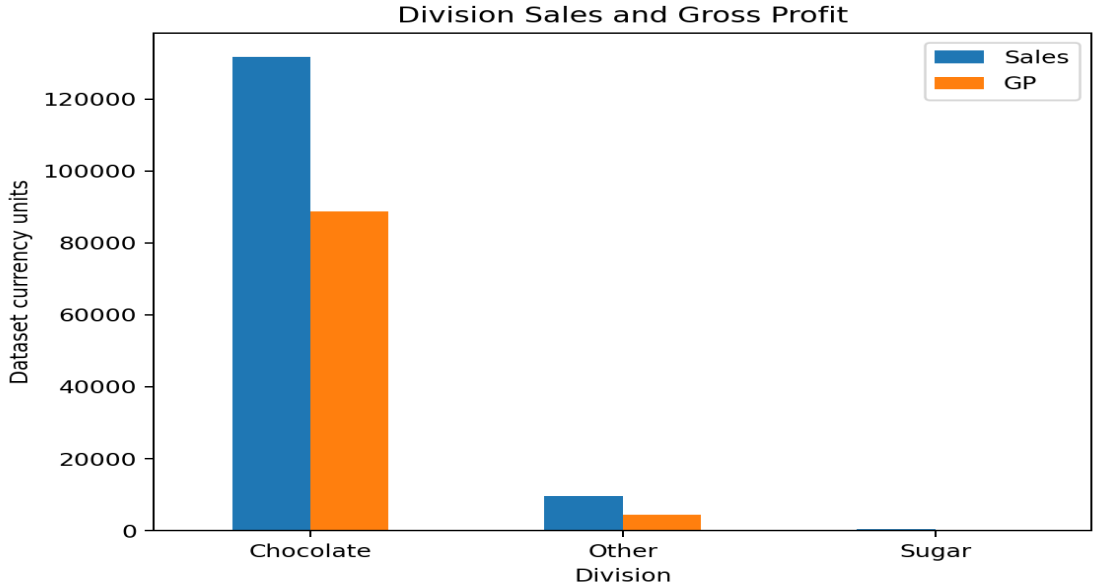
Product	Sales	Gross profit	Margin	Profit share
Wonka Bar -Scrumdiddlyumptious	27,874.80	19,357.50	69.44%	20.72%
Wonka Bar - Triple Dazzle Caramel	28,485.00	18,610.20	65.33%	19.92%
Wonka Bar - Milk Chocolate	26,867.75	17,443.37	64.92%	18.67%
Wonka Bar - Nutty Crunch Surprise	23,574.95	16,819.95	71.35%	18.00%
Wonka Bar - Fudge Mallows	24,890.40	16,593.60	66.67%	17.76%
Lickable Wallpaper	7,860.00	3,930.00	50.00%	4.21%
Wonka Gum	597.50	310.70	52.00%	0.33%
Everlasting Gobstopper	130.00	104.00	80.00%	0.11%



Wonka Bar -Scrumdiddlyumptious leads gross profit with 19,357.50 and 20.72% of total profit. The five leading Wonka Bar products generate 95.06% of gross profit. Kazookles has the lowest margin at 7.69% on 1,205.75 sales; this merits cost and pricing review but does not alone justify discontinuation.

6.3 Division performance

Division	Sales	Gross profit	Margin	Revenue share	Profit share
Chocolate	131,692.90	88,824.62	67.45%	92.88%	95.06%
Other	9,663.25	4,333.45	44.84%	6.82%	4.64%
Sugar	427.48	284.73	66.61%	0.30%	0.30%

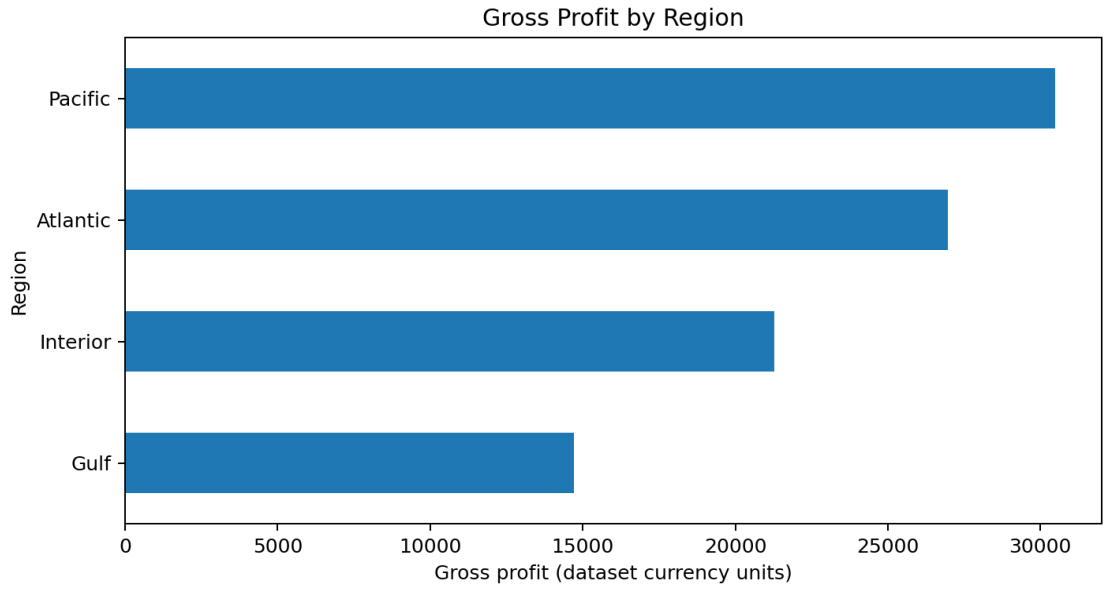


Chocolate contributes 95.06% of gross profit from 92.88% of revenue. Other has a weighted margin of 44.84%, below Chocolate.

6.4 Regional and state performance

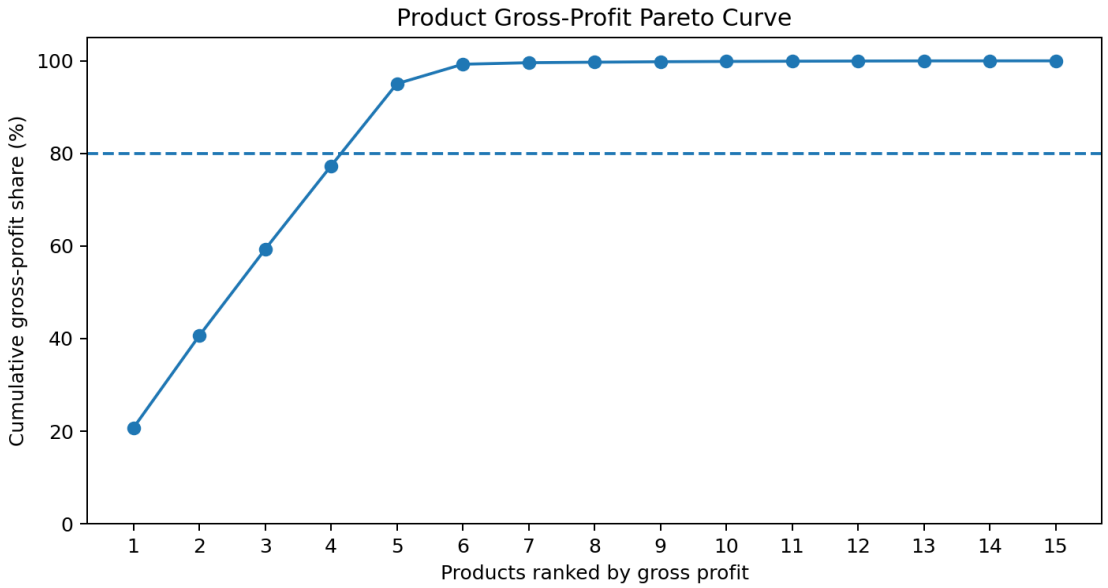
Region	Sales	Gross profit	Margin	Profit share
Pacific	46,301.53	30,485.94	65.84%	32.63%

Region	Sales	Gross profit	Margin	Profit share
Atlantic	41,197.24	26,973.70	65.47%	28.87%
Interior	32,037.60	21,282.49	66.43%	22.78%
Gulf	22,247.26	14,700.67	66.08%	15.73%



Pacific leads regional gross profit at 30,485.94 (32.63% of total). Regional margins range from 65.47% to 66.43%. The top five states/provinces contribute 50.73% of gross profit.

6.5 Pareto and HHI



Four leading products account for 77.30% of gross profit; five account for 95.06%. The profit HHI is 1,831; revenue HHI is 1,766. The results indicate concentration in the Wonka Bar family.

6.6 RFM segmentation

Segment	Customers	Customer share	Sales	Revenue share	Avg recency	Avg orders
At Risk / Low Value	1,103	21.87%	11,647.02	8.21%	450	1.01

Segment	Customers	Customer share	Sales	Revenue share	Avg recency	Avg orders
Developing	1,776	35.21%	30,927.34	21.81%	266	1.23
Loyal	1,489	29.52%	57,859.50	40.81%	279	2.16
Champions	676	13.40%	41,349.77	29.16%	111	3.02

2,811 customers (55.7%) placed one order. RFM groups are descriptive; alternative cut-offs, reference dates and tie handling can produce different segment totals.

7. Discussion

Profit is concentrated in five Wonka Bar products, creating product-family dependency. Low-margin products should be reviewed in the context of sales scale and strategic role. Regional margins vary within a narrow band, so regional gross-profit differences are more consistent with sales scale and mix than with large margin differences. RFM can prioritise customer engagement, but the dataset cannot explain inactivity or prove that a campaign will bring customers back.

8. Management Recommendations

• Protect availability, quality and supplier continuity for the five leading profit contributors.

• Review Kazookles pricing and supplier cost before changing its assortment role.

• Compare profit share with revenue share when reviewing product and division performance.

• Use regional sales, margin and product mix together in commercial planning.

• Test customer re-engagement using a comparison group and measure incremental orders and gross profit.

• Verify currency, investigate the 313 Row-ID-only duplicates, correct Order ID year tokens and obtain an authoritative factory mapping.

9. Limitations

• Gross profit is not net profit; unrecorded overhead, freight, marketing, tax and financing are not included.

• Currency is not independently verified.

• Ship Date is unreliable and excluded.

• 313 rows identical except Row ID are retained pending source validation.

• Factory mapping is absent.

• Some products have very low order counts.

• RFM labels and cut-offs are analyst-defined.

• The observational data do not establish causality or forecast future behaviour.

10. Conclusion

The study converts 10,194 transaction lines into a reproducible profitability and customer analytics framework. Sales total 141,783.63 and gross profit 93,442.80, with a weighted gross margin of 65.91%. Five Wonka Bar products account for 95.1% of gross profit; Chocolate is the leading division and Pacific the leading region. Priorities include protecting core profit contributors, reviewing weak-margin products, improving data governance and testing customer-retention actions.

11. Future Scope

Future work can add inventory, stockout, supplier, freight, operating-expense and product price-history data. Forecasting and inventory optimisation require longer validated time series. Churn modelling requires a defined outcome and temporal validation. Retention strategies should be tested through controlled experiments.

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Appendix A. Reproducibility

Primary input: Nassau_Candy_Distributor.csv. Results were recalculated using pandas. Order Date was parsed as dd-mm-yyyy; gross profit was recalculated as Sales – Cost; weighted margins use aggregate ratios. RFM reference date: 1 January 2026; quintile scoring uses rank(method="first"). The figures are in dataset currency units. Dashboard outputs may differ if its filters or segmentation rules differ from this paper.